

Project 2: Numerical Solutions to Incompressible Navier–Stokes Equations

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Write a concise abstract here. State the governing equations solved, the numerical scheme implemented (fractional-step method with second-order Adams–Bashforth advection and Crank–Nicolson diffusion on a staggered grid), the Reynolds numbers investigated, and key findings regarding accuracy and physical behavior of the lid-driven cavity flow.

I. INTRODUCTION

Introduce the lid-driven cavity problem and its role as a standard benchmark in incompressible CFD[?]. Briefly explain why it is nontrivial (recirculating flow, corner vortices, Re -dependent topology) and what makes the fractional-step method[?] an effective approach for incompressible flows.

II. GOVERNING EQUATIONS

The dimensionless 2-D incompressible Navier–Stokes equations are

$$\frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} \cdot \nabla \mathbf{q} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{q}, \quad (1)$$

$$\nabla \cdot \mathbf{q} = 0, \quad (2)$$

where $\mathbf{q} = \{u, v\}$ is the velocity field, p is pressure, and Re is the Reynolds number. Describe the boundary conditions (no-slip on three walls; lid velocity $u_{\text{top}} = 1$ on the top boundary).

III. OPERATORS

III.A. Gradient Operator

The gradient operator ($\mathbf{G}$) is discretized using central differences for the interior points and one-sided differences for the boundary points. It maps pressure difference to velocity face values using the following formulas:

$$(\mathbf{G}p)_{i,j}^u = \frac{p_{i,j} - p_{i-1,j}}{\Delta x}, \quad (\mathbf{G}p)_{i,j}^v = \frac{p_{i,j} - p_{i,j-1}}{\Delta y} \quad (3)$$

III.B. Divergence Operators

The divergence operator ($\mathbf{D}$) is discretized using central differences for the interior points and one-sided differences for the boundary points. It maps velocity face values to cell-centered divergence. The central points are calculated using the following formula:

$$(\mathbf{D}u)_{i,j} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} + \frac{v_{i,j+1} - v_{i,j}}{\Delta y} \quad (4)$$

The full divergence operator is split into the sum of the interior and boundary contributions. The boundary contribution ($\mathbf{bc}_D$) at each wall is given in the following table:

FIG. 1 Staggered grid formulation. Horizontal velocity u ($\rightarrow$), vertical velocity v ($\uparrow$), and pressure p ($\blacksquare$) are located at cell faces and cell centers, respectively. The open circle ($\circ$) marks the pinned pressure reference.

III.C. Laplacian Operators

For the interior points:

$$(\mathbf{L}u)_{i,j} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2} \quad (5)$$

III.D. Advection Operator

III.E. Unit Testing of Operators

IV. NUMERICAL METHOD

IV.A. Staggered Grid Formulation

Describe the staggered grid layout with velocity components defined at cell faces and pressure at cell centers. State the grid dimensions for each variable and the pressure reference condition.

IV.B. Spatial Discretization

Introduce the discrete divergence (D), gradient (G), advection (N), and Laplacian (L) operators. State their orders of accuracy and discretization stencils.

IV.C. Time Integration: Fractional-Step Method

Using second-order Adams–Bashforth for advection and Crank–Nicolson for diffusion, the semi-discrete momentum equation and continuity constraint form the saddle-point system[?]

$$\begin{pmatrix} A & G \\ D & 0 \end{pmatrix} \begin{pmatrix} \mathbf{q}^{n+1} \\ p^{n+1} \end{pmatrix} = \begin{pmatrix} \mathbf{r}^n + \mathbf{bc}_1 \\ 0 \end{pmatrix}, \quad (6)$$

where $A = I/\Delta t - L/(2Re)$ and $\mathbf{r}^n$ contains the explicit Adams–Bashforth advection and Crank–Nicolson diffusion terms. Applying block LU decomposition and approximating $A^{-1} \approx \Delta t I$ yields the three-step fractional-

TABLE I Unit tests for discrete operators, verifying correctness of each component.

Test	Operator	What is checked	Expected result	Pass?
G1	G	Gradient of linear p	Exact (machine ε)	✓
G2	G	Gradient of quadratic p	Error $O(\Delta x^2)$	✓
D1	DG	Composition $\approx \nabla^2$	Error $O(\Delta x^2)$ interior	✓
D2	D	Divergence of div-free field	Error $O(\Delta x^2)$ interior	✓
D3	D, G	Adjoint: $\mathbf{u}^\top \mathbf{G} \mathbf{p} = \mathbf{p}^\top \mathbf{D} \mathbf{u}$	Machine ε	✓
BC1	bc_D	Global mass balance $\sum \mathbf{bc}_D = 0$	Exact	✓
BC2	bc_L	Top-wall BC value $= 2U^*/\Delta y^2$	Exact	✓
L1	L	Laplacian of quadratic field $= 2$	Exact interior	✓
L2	L	Laplacian of linear field $= 0$	Machine ε	✓
L3	L	Self-adjoint: $\mathbf{u}^\top \mathbf{L} \mathbf{v} = \mathbf{v}^\top \mathbf{L} \mathbf{u}$	Machine ε	✓
A1	A	Uniform flow gives zero advection	Machine ε	✓
A2	A	Energy: $\mathbf{u}^\top \mathbf{A}(\mathbf{u}) \approx 0$	Small ($O(\Delta x^2)$)	✓
CG1	CG	Solves Poisson problem to tolerance	$< \text{tol}$	✓
FULL	T	$\mathbf{p}^\top \mathbf{T} \mathbf{p} > 0$ (SPD)	Positive	✓

FIG. 2 (a) Temporal convergence versus CFL and (b) spatial convergence versus grid size N for v at the cavity center. Reference lines indicate the expected convergence rates.FIG. 3 (a) Horizontal velocity u along the vertical centerline $x = 0.5$, (b) vertical velocity v along the horizontal centerline $y = 0.5$, and (c) vorticity ω along the top wall $y = 1.0$ for several Reynolds numbers. Solid lines: present simulation; symbols: Ghia et al.[?].

step algorithm:

$$A\mathbf{q}^* = \mathbf{r}^n + \text{bc}_1, \quad (7)$$

$$D \Delta t G p^{n+1} = D\mathbf{q}^*, \quad (8)$$

$$\mathbf{q}^{n+1} = \mathbf{q}^* - \Delta t G p^{n+1}. \quad (9)$$

The intermediate velocity system (7) and pressure Poisson equation (8) are solved at each time step using a conjugate gradient method[?].

IV.D. Time Step and Stopping Criterion

The time step is chosen as

$$\Delta t = \min\left(\text{CFL} \frac{\Delta x}{u_{\text{top}}}, \text{CFL} (\Delta x)^2 Re\right), \quad (10)$$

and time integration proceeds until the steady-state criterion $\max |\mathbf{q}^{n+1} - \mathbf{q}^n| < 10^{-8}$ is satisfied.

V. RESULTS

V.A. Grid and Temporal Convergence

Present log-log convergence plots for vertical velocity v at the cavity center $(x, y) = (0.5, 0.5)$.

V.B. Validation Against Ghia et al.

Figure 3 compares centerline velocities and top-wall vorticity to the reference data of Ghia, Ghia, and Shin[?] for $Re = \{100, 400, 1000, 3200, 5000\}$.

FIG. 4 Streamline contours at $Re = 100$ (top), 1000 (middle), and 3200 (bottom). Left column: present simulation; right column: Ghia et al.[?].

V.C. Streamline Topology

Figure 4 shows streamline patterns for $Re = \{100, 1000, 3200\}$. Discuss the evolution of the primary vortex center and the growth of secondary corner vortices with increasing Reynolds number.

V.D. Vorticity Contours

Present vorticity contours and compare qualitatively and quantitatively to the reference[?]. Discuss boundary-layer thinning at higher Re and any limitations of the uniform-grid approach.

V.E. Additional Exploration

Present additional results that go beyond the required validation. For example, ...

VI. DISCUSSION

Discuss the observed convergence rates in light of the numerical scheme, the accuracy of results at varying Reynolds numbers, and physical observations from the flow field. Address any discrepancies with the reference data.

VII. CONCLUSION

Summarize the key findings of the project in 4–6 sentences. State clearly the validated Reynolds number range, convergence rates observed, and any limitations identified.

NOTE ON CODE SUBMISSION

Your MATLAB/Python code should be appended *after* the references as a separate PDF printout. It does

not count toward the 10-page limit and does not need to appear in the L^AT_EX source.